

Dataset Description

1. Overview

The **VinDr-CXR** (VinBigData Chest X-ray) dataset was created for the automatic detection and localization of thoracic abnormalities in chest radiographs. It was released by VinBigData in collaboration with **Hospital 108** and **Hanoi Medical University Hospital** in Vietnam.

This dataset is unique because it provides **radiologist-level annotations** for both classification (presence of disease) and localization (bounding boxes), tackling the challenge of label noise and observer variability common in medical imaging.

2. Dataset Composition

The dataset consists of **18,000** postero-anterior (PA) view chest X-ray scans.

- **Training Set:** 15,000 images (Publicly labeled).
- **Test Set:** 3,000 images (Labels withheld for competition/benchmarking).
- **Image Format:** DICOM (.dicom or .dcm).
- **Bit Depth:** High dynamic range (often 12-bit or 16-bit), requiring normalization to 8-bit for standard computer vision models.

Training Data (15,000 images)

- **Independent Labeling:** Each image in the training set was independently annotated by **3 experienced radiologists**.
- **Result:** This leads to multiple annotations for the same image. For example, one radiologist might mark a "Nodule," while another might mark nothing or a different location.
- **Implication for Models:** Models must handle noisy or conflicting ground truth labels (e.g., using Weighted Box Fusion or majority voting).

Test Data (3,000 images)

- **Consensus Labeling:** Each image in the test set was labeled by the **consensus of 5 radiologists** to ensure a high-quality "Gold Standard" for evaluation.

4. Class Labels and Statistics

The dataset focuses on **14 critical radiographic findings**. The train.csv file contains **67,914** total annotations across the 15,000 training images.

Class ID Map

Class ID	Class Name	Count (Train)	Description
0	Aortic enlargement	7,162	Abnormal widening of the aorta.
1	Atelectasis	279	Collapse of lung tissue.
2	Calcification	960	Calcium deposits in the lung.
3	Cardiomegaly	5,427	Enlarged heart.
4	Consolidation	556	Lung tissue filled with liquid instead of air.
5	ILD	1,000	Interstitial Lung Disease.
6	Infiltration	1,247	Substance denser than air (pus, blood) in lungs.
7	Lung Opacity	2,483	Opaque areas in the lung.
8	Nodule/Mass	2,580	Growths or lumps in the lung.
9	Other lesion	2,203	Lesions not fitting other categories.
10	Pleural effusion	2,476	Fluid build-up between lung layers.
11	Pleural thickening	4,842	Thickening of the pleura.
12	Pneumothorax	226	Air leaking into the space between

			lung and chest wall.
13	Pulmonary fibrosis	4,655	Scarring of the lung tissue.
14	No finding	31,818	No abnormalities detected.

Distribution Analysis

- **Dominant Class:** "No finding" (Class 14) is the most frequent, representing ~47% of all annotations.
- **Rare Classes:** *Pneumothorax* (226), *Atelectasis* (279), and *Consolidation* (556) are rare, making the dataset highly imbalanced. Techniques like oversampling or class-weighted loss functions are recommended.

5. File Structure and Metadata

train.csv Structure

The annotation file contains the following columns:

- `image_id`: Unique identifier for the scan.
- `class_name`: Human-readable name of the abnormality (e.g., "Cardiomegaly").
- `class_id`: Integer ID mapping to the class name (0-14).
- `rad_id`: Identifier of the radiologist who made the annotation (e.g., R10, R9). This allows filtering by specific annotators.
- `x_min`, `y_min`, `x_max`, `y_max`: Coordinates of the bounding box.
 - **Note:** For "No finding" (Class 14), these values are NaN.

Directory Structure

```
/kaggle/input/vinbigdata-chest-xray-abnormalities-detection/
|
├── train.csv          # Annotations for training images
├── sample_submission.csv # Format for submission
├── train/             # Folder containing 15,000 DICOM images
│   ├── [image_id].dicom
│   └── ...
└── test/              # Folder containing 3,000 DICOM images
    ├── [image_id].dicom
    └── ...
```

6. Preprocessing Requirements

As identified in the analysis notebook (image-conversion-and-annotation.ipynb), the raw data requires specific handling:

1. **DICOM Handling:** Files are in DICOM format, containing pixel data and metadata.
 - **Photometric Interpretation:** Some images may look inverted (MONOCHROME1 vs MONOCHROME2) and need to be corrected.
2. **Normalization:** Pixel arrays often contain high values (e.g., > 2000). To use standard CNNs (ResNet, EfficientNet, YOLO), images must be:
 - Normalized (min-max scaling).
 - Converted to 8-bit integers (0-255).
 - Saved as standard formats like PNG or JPG.
3. **Resize:** Original DICOMs are very high resolution (often 2000x2000+). Resizing to standard input sizes (e.g., 512x512 or 1024x1024) is necessary for training efficiency.